The numerical toolbox II: matrices and waves

Prognostic Modelling (pea700) · Lecture 4

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Recap of lecture 3

- Finite differences from the Taylor series; order of convergence p: halving h divides the error by 2^p
- Round-off limits how small h can be
- Stencils of any order from a small linear system
- Quadrature and root finding: a nonlinear problem becomes a sequence of **linear systems**

Today

1. **The model matrix** A of $-u'' = f$: sparse, symmetric, positive definite → exactly one solution
2. **Its eigenvalues** λ_m as formulas → needed for 3 and 4
3. **Norm and condition number**: $\|A\|_2 = \lambda_{\max}, \kappa = \lambda_{\max}/\lambda_{\min}$ → how errors grow
4. **Waves on a grid**: DFT, aliasing, modified wavenumber → the eigenvalues once more

> Two views of one discretization: the matrix acts on all grid points at once; waves show what it does to each scale. The eigenvectors of the matrix **are** waves.

From a differential equation to a matrix

A model problem

$$-\frac{\mathrm{d}^2 u}{\mathrm{d}x^2} = f(x), \qquad u(0) = u(L) = 0$$

$N+1$ intervals, $h = L/(N+1)$, $x_j = jh$; known $u_0 = u_{N+1} = 0$, unknowns $u_1, \dots, u_N$.
Central difference at each interior point:

$$\frac{-u_{j-1} + 2u_j - u_{j+1}}{h^2} = f_j, \qquad j = 1, \dots, N$$

Multiply by h^2; for $N = 4$, with $u_0 = u_5 = 0$ in the first and last row:

$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix} = h^2 \begin{pmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{pmatrix}$$

The model matrix

$$A\vec{x} = \vec{b}, \qquad \vec{x} = \begin{pmatrix} u_1 \\ \vdots \\ u_N \end{pmatrix}, \qquad \vec{b} = h^2 \begin{pmatrix} f_1 \\ \vdots \\ f_N \end{pmatrix}, \qquad A = \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix}$$

The h^2 stands on the **right** hand side, because we multiplied each equation by it.

- Row j = the stencil $(-1, 2, -1)$, centred on the diagonal
- A accompanies us through the elliptic and parabolic units

Think first: how many of the N^2 entries are non-zero?

$3N - 2$: the matrix is **sparse**. With 10^8 grid points the full matrix would have 10^{16} entries; stored without zeros (`scipy.sparse`) it has $3 \cdot 10^8$.

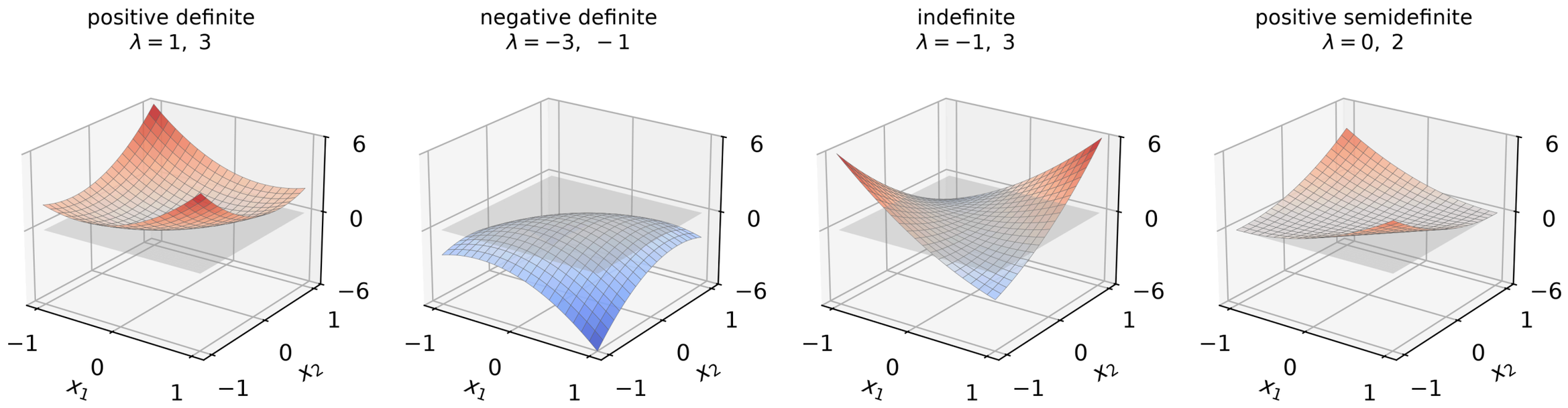
Symmetry and definiteness

Symmetric, $A = A^T$: the stencil couples to the left neighbour with the same weight, -1, as to the right.

Definiteness = the sign of the **quadratic form** $q(\vec{x}) = \vec{x}^T A \vec{x} = \sum_i \sum_j x_i \, a_{ij} \, x_j$:

$q(\vec{x})$ **for all** $\vec{x} \neq 0$	**the matrix is**
> 0	positive definite
≥ 0, and $= 0$ for some $\vec{x} \neq 0$	positive semidefinite
< 0	negative definite
≤ 0, and $= 0$ for some $\vec{x} \neq 0$	negative semidefinite
> 0 for some $\vec{x}$, < 0 for others	indefinite

Four quadratic forms



Bowl, upside-down bowl, saddle, trough. Grey: the plane $q = 0$. Left: the model matrix for $N = 2$, $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$; right: $\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$, $q = (x_1 - x_2)^2$, zero along $x_1 = x_2$.

Test 1: a sum of squares

$N = 3$, multiply out row by row:

$$\begin{aligned}\vec{x}^T A\vec{x} &= x_1(2x_1 - x_2) + x_2(-x_1 + 2x_2 - x_3) + x_3(-x_2 + 2x_3) \\ &= 2x_1^2 + 2x_2^2 + 2x_3^2 - 2x_1x_2 - 2x_2x_3\end{aligned}$$

Split each $2x_j^2 = x_j^2 + x_j^2$; each mixed term completes a binomial:

$$= x_1^2 + \underbrace{(x_1^2 - 2x_1x_2 + x_2^2)}_{(x_1-x_2)^2} + \underbrace{(x_2^2 - 2x_2x_3 + x_3^2)}_{(x_2-x_3)^2} + x_3^2$$

For every N: $\vec{x}^T A\vec{x} = x_1^2 + \sum_{j=1}^{N-1}(x_j - x_{j+1})^2 + x_N^2 \geq 0, \quad = 0$ only for $\vec{x} = 0$: **positive definite**.

x_1^2, x_N^2 come from the Dirichlet values $u_0 = u_{N+1} = 0$. Without them (Neumann, periodic) a constant $\vec{x}$ gives $q = 0$: only **semidefinite**.

Test 2: the eigenvalues

Eigenvector $\vec{v} \neq 0$, eigenvalue λ: $A\vec{v} = \lambda\vec{v}$. Symmetric A: real λ, N orthonormal eigenvectors, $\vec{v}_m^T \vec{v}_n = 1$ if $m = n$, else 0.

Write $\vec{x} = \sum_m c_m \vec{v}_m$. Then $A\vec{x} = \sum_n c_n \lambda_n \vec{v}_n$ and only $m = n$ survives:

$$\vec{x}^T A \vec{x} = \sum_m \sum_n c_m c_n \lambda_n \, \vec{v}_m^T \vec{v}_n = \sum_m \lambda_m c_m^2$$

All $\lambda_m > 0$: positive definite. One $\lambda_m < 0$: $\vec{x} = \vec{v}_m$ gives $q = \lambda_m < 0$.

$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$, $\lambda = 3, -1$: $\vec{x} = (1, 1)$ gives $q = 6$, $\vec{x} = (1, -1)$ gives $q = -2$: the saddle.

For A itself we need its eigenvalues: next section.

Why we want positive definite

Positive definite ⇒ **invertible**: if $A\vec{x} = \vec{0}$ for some $\vec{x} \neq \vec{0}$, then $\vec{x}^T A\vec{x} = 0$, which is not allowed.

So $A\vec{x} = \vec{b}$ has **exactly one solution** for every $\vec{b}$: the discrete problem is well-posed.

Neumann or periodic boundaries: the constant vector gives $A\vec{x} = \vec{0}$, the solution is fixed only up to a constant.

The sign matters. The second difference $(u_{j-1} - 2u_j + u_{j+1})/h^2$ has the matrix $-A/h^2$: **negative** definite. That is why we write $-u'' = f$.

> **Result.** The model matrix is symmetric positive definite (SPD): $\vec{x}^T A\vec{x} = x_1^2 + \sum_j (x_j - x_{j+1})^2 + x_N^2 > 0$. Exactly one solution. The fastest solvers need SPD (unit on iterative solvers).

Eigenvalues of the model matrix

The goal

Formulas for all N eigenvalues λ_m and eigenvectors $\vec{v}_m$ of A, in three steps:

1. Guess the eigenvectors from the continuous problem
2. The interior rows give the eigenvalue
3. The boundary rows fix which waves are allowed

We need the result three times:

- definiteness: are all $\lambda_m > 0$?
- norm and condition number: $\lambda_{\max}$ and $\lambda_{\min}$
- the link to waves and Fourier analysis

Step 1: which problem do the eigenvectors solve?

Not the Poisson equation: u depends on f, an eigenvector does not. The continuous counterpart of $A\vec{v} = \lambda\vec{v}$:

$$-\frac{\mathrm{d}^2 v}{\mathrm{d}x^2} = \mu\, v, \qquad v(0) = v(L) = 0$$

$v = \sin(kx)$: $-v'' = k^2 \sin(kx)$, so $\mu = k^2$; $v(L) = 0$ requires $k = m\pi/L$: m **half waves** in the domain.

Think first: A is $-\mathrm{d}^2/\mathrm{d}x^2$ times h^2. Guess its eigenvectors.

The same sines on the grid: $v_j = \sin(j\theta)$, with the **phase step per grid point**

$$\theta = kh \qquad (\text{wavenumber} \times \text{grid spacing, no unit})$$

Long wave: θ small. Shortest wave on the grid: θ close to π.

Step 2: the interior rows give the eigenvalue

$$(A\vec{v})_j = -\sin\Big((j-1)\theta\Big) + 2\sin(j\theta) - \sin\Big((j+1)\theta\Big)$$

Add $\sin(a \mp b) = \sin a \cos b \mp \cos a \sin b$: $\sin(a-b) + \sin(a+b) = 2\sin a \cos b$, with $a = j\theta, b = \theta$:

$$(A\vec{v})_j = 2\sin(j\theta) - 2\sin(j\theta)\cos\theta = (2 - 2\cos\theta)\, v_j$$

Every row gives the same factor: the eigenvalue is $2 - 2\cos\theta$, for any θ so far.

Step 3: the boundary rows fix θ

First and last row lack a neighbour. The formula holds there only if the missing values vanish, like $u_0 = u_{N+1} = 0$ at the boundary nodes $j = 0$ and $j = N + 1$:

- left: $v_0 = \sin 0 = 0$ automatically
- right: $v_{N+1} = \sin\Big((N+1)\theta\Big) = 0$ requires $(N+1)\theta = m\pi$

$m = 1, \dots, N$ give N different eigenvectors ($m = 0, N+1$ give $\vec{0}$; larger m repeat): all of them.

Check with the continuous problem: $h = L/(N+1)$, so $\theta = kh = \dfrac{m\pi}{L} \cdot \dfrac{L}{N+1} = \dfrac{m\pi}{N+1}$ ✓

Result: eigenvalues of the model matrix

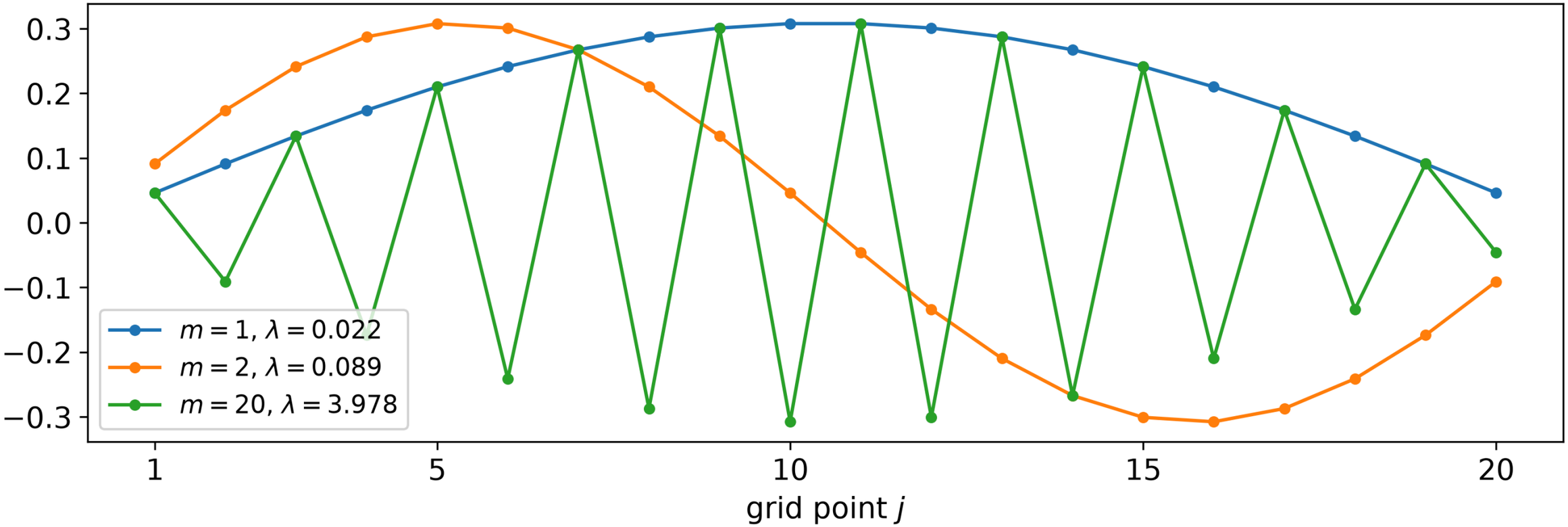
For $m = 1, 2, \dots, N$:

$$\theta_m = \frac{m\pi}{N+1}, \qquad \lambda_m = 2 - 2\cos\theta_m = 4\sin^2\frac{\theta_m}{2}, \qquad (\vec{v}_m)_j = \sqrt{\tfrac{2}{N+1}}\,\sin(j\theta_m)$$

- m = number of **half waves**; $m = 1$ longest, $m = N$ shortest wave
- $\theta_m = k_m h$: phase step per grid point; $\sqrt{2/(N+1)}$ scales $\vec{v}_m$ to length 1
- $0 < \theta_m < \pi$, so $0 < \lambda_1 < \lambda_2 < \cdots < \lambda_N < 4$

> $\lambda_{\min} = \lambda_1, \lambda_{\max} = \lambda_N$. All $\lambda_m > 0$: positive definite, as test 1 showed.

Eigenvectors are sine waves



m half waves across the domain. Smallest λ: longest wave; largest λ: shortest wave.

What the eigenvectors are good for

Write the right hand side in eigenvectors, $\vec{b} = \sum_m \beta_m \vec{v}_m$ with $\beta_m = \vec{v}_m^T \vec{b}$. Then

$$\vec{u} = \sum_m \frac{\beta_m}{\lambda_m} \vec{v}_m \qquad \text{solves} \qquad A\vec{u} = \vec{b},$$

because A multiplies each $\vec{v}_m$ by λ_m and restores β_m.

- Each wave of the right hand side is **divided by its eigenvalue**: the discrete Fourier sine series
- Long waves, $\lambda_m \approx \theta_m^2 = (kh)^2$: with $\vec{b} = h^2 \vec{f}$ the factor is $h^2/\lambda_m \approx 1/k^2 = 1/\mu$, as in the continuous problem
- Errors in $\vec{b}$ are divided by λ_m too: most amplified in $m = 1$ (by $1/\lambda_1$), least in $m = N$ (by $1/\lambda_N$)

> The ratio λ_N/λ_1 measures how much a linear system can amplify errors: the **condition number**, next.

The condition number

The question

If $\vec{b}$ contains an error (round-off, measurement), how large is the error in the solution of $A\vec{x} = \vec{b}$?

1. Measure vectors and matrices: **norms**
2. Bound the error: the **condition number**
3. Evaluate it for the model matrix, with λ_1 and λ_N

Norms

$$\|\vec{x}\|_2 = \sqrt{\sum_j x_j^2}, \qquad \|\vec{x}\|_\infty = \max_j |x_j|$$

Norms are what we mean by “the error” on a grid. Matrix norm: the largest stretching factor,

$$\|A\| = \max_{\vec{x}\neq 0} \frac{\|A\vec{x}\|}{\|\vec{x}\|} \qquad \Longrightarrow \qquad \|A\vec{x}\| \leq \|A\|\,\|\vec{x}\|$$

How large is $\|A\|_2$? For a symmetric matrix the eigenvalues tell us: next slide.

The 2-norm of a symmetric matrix

Use the orthonormal eigenvectors: $\vec{x} = \sum_m c_m \vec{v}_m$, so $\|\vec{x}\|_2^2 = \sum_m c_m^2$ and $A\vec{x} = \sum_m \lambda_m c_m \vec{v}_m$:

$$\|A\vec{x}\|_2^2 = \sum_m \lambda_m^2 c_m^2 \leq \max_m \lambda_m^2 \sum_m c_m^2 = \max_m \lambda_m^2 \, \|\vec{x}\|_2^2$$

Each part is stretched by its own eigenvalue; equality for $\vec{x}$ = the eigenvector with the largest $|\lambda_m|$.

A^{-1}: same eigenvectors, eigenvalues $1/\lambda_m$ (from $A\vec{v}_m = \lambda_m \vec{v}_m$).

> **Result.** $\|A\|_2 = \max_m |\lambda_m|$, $\|A^{-1}\|_2 = 1/\min_m |\lambda_m|$. Model matrix: $\|A\|_2 = \lambda_N$, $\|A^{-1}\|_2 = 1/\lambda_1$.

How errors are amplified

Perturb the right hand side, $A(\vec{x} + \delta\vec{x}) = \vec{b} + \delta\vec{b}$; subtract $A\vec{x} = \vec{b}$:

$$A\,\delta\vec{x} = \delta\vec{b} \quad \Longrightarrow \quad \delta\vec{x} = A^{-1}\delta\vec{b} \quad \Longrightarrow \quad \|\delta\vec{x}\| \le \|A^{-1}\|\,\|\delta\vec{b}\|$$

And from $\vec{b} = A\vec{x}$: $\|\vec{b}\| \le \|A\|\,\|\vec{x}\|$, i.e. $1/\|\vec{x}\| \le \|A\|/\|\vec{b}\|$. Multiply:

$$\frac{\|\delta\vec{x}\|}{\|\vec{x}\|} \le \underbrace{\|A\|\,\|A^{-1}\|}_{\kappa(A)}\,\frac{\|\delta\vec{b}\|}{\|\vec{b}\|}$$

Result. With the 2-norm result, an SPD matrix has $\kappa(A) = \lambda_{\max}/\lambda_{\min}$. Rule of thumb: $\kappa = 10^{p}$ costs about p of the 16 digits.

The condition number of the model matrix

Insert the eigenvalues. Largest, $\lambda_{\max} = \lambda_N$: $\theta_N = N\pi/(N+1) \approx \pi$, $\cos\theta_N \approx -1$, so $\lambda_N \approx 4$.

Smallest, $\lambda_{\min} = \lambda_1$: $\theta_1 = \pi/(N+1)$ small, $\cos\theta \approx 1 - \theta^2/2$:

$$\lambda_1 = 2 - 2\cos\theta_1 \approx \theta_1^2 = \frac{\pi^2}{(N+1)^2} \quad \Longrightarrow \quad \kappa(A) = \frac{\lambda_N}{\lambda_1} \approx \frac{4(N+1)^2}{\pi^2} \approx 0.4\,N^2$$

```
N =   10: condition number      48.4,  estimate      49.0
N =  100: condition number    4133.6,  estimate    4134.3
N = 1000: condition number  406095.0,  estimate  406095.7
```

Result. $\kappa \sim N^2 \sim 1/h^2$; $N = 1000$: $\kappa \approx 4 \times 10^5$, about 6 digits lost. A finer grid is more accurate **and** harder to solve: two units on linear solvers.

Fourier analysis

Three questions

The eigenvectors of A are sine waves. Turn this into a general tool:

1. How do we split any grid function into waves? → **DFT**
2. Which waves can a grid represent at all? → **aliasing**
3. What does a difference formula do to each wave? → **modified wavenumber**, and the eigenvalues λ_m once more

The tool for all three: the **complex exponential**.

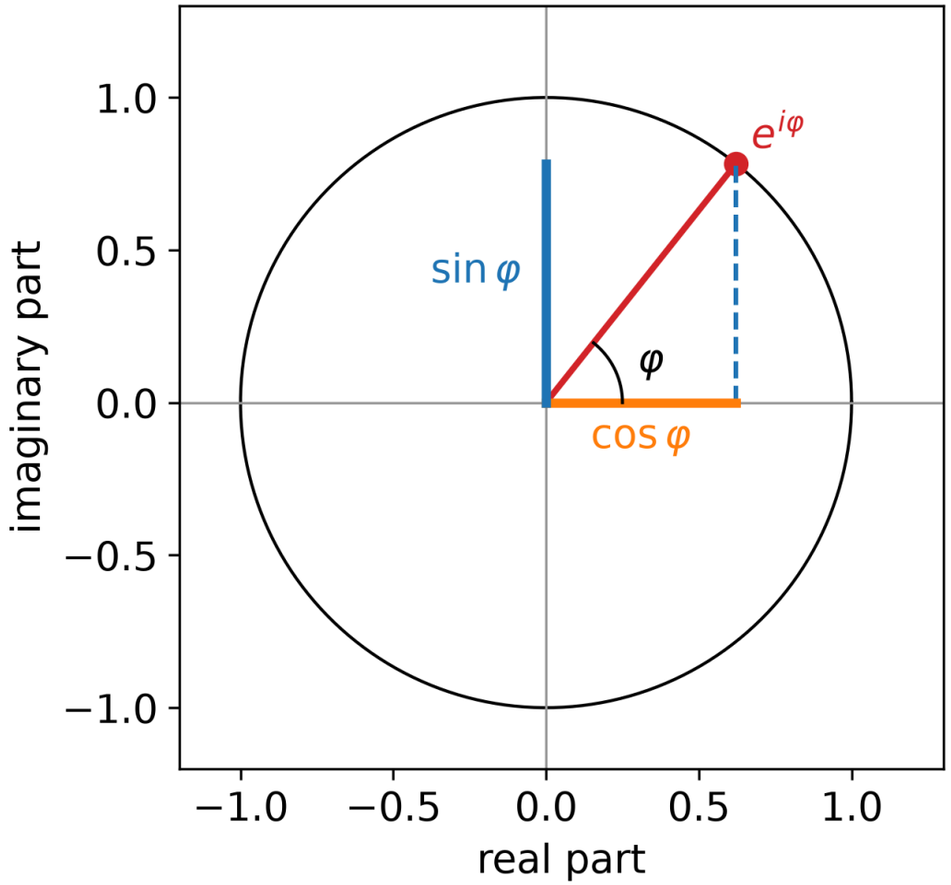
The complex exponential

$$e^{i\varphi} = \cos\varphi + i\sin\varphi$$

A point on the unit circle at angle φ: real part $\cos$, imaginary part $\sin$. Only new rule: $i^2 = -1$.

Taylor series of e^x, with $i^2 = -1, i^3 = -i$:

$$\begin{aligned} e^{i\varphi} &= 1 + i\varphi - \frac{\varphi^2}{2} - i\frac{\varphi^3}{6} + \dots \\ &= \left(1 - \frac{\varphi^2}{2} + \dots\right) + i\left(\varphi - \frac{\varphi^3}{6} + \dots\right) \end{aligned}$$



Why it makes life easier

- **Multiplying adds angles:** $e^{ia}e^{ib} = e^{i(a+b)}$, all addition theorems in one line
- **A shift is a multiplication:** $e^{ik(x+\Delta x)} = e^{ikx}\, e^{ik\Delta x}$
- **A derivative is a multiplication:** $\frac{\mathrm{d}}{\mathrm{d}x} e^{ikx} = ik\, e^{ikx}$, $\frac{\mathrm{d}^2}{\mathrm{d}x^2} e^{ikx} = -k^2 e^{ikx}$
- **Real waves** are two opposite rotations: $\cos kx = \frac{1}{2}\left(e^{ikx} + e^{-ikx}\right)$

The model matrix once more, with $v_j = e^{ij\theta}$, no addition theorem needed:

$$(A\vec{v})_j = -e^{i(j-1)\theta} + 2e^{ij\theta} - e^{i(j+1)\theta} = e^{ij\theta}\left(2 - e^{-i\theta} - e^{i\theta}\right) = (2 - 2\cos\theta)\, v_j$$

using $e^{i\theta} + e^{-i\theta} = 2\cos\theta$.

Back to the sine

$e^{ij\theta} = \cos(j\theta) + i\sin(j\theta)$ is a trick for the algebra, not a new eigenvector.

A is real, so it acts on real and imaginary part separately. Compare both sides of $A\vec{v} = (2 - 2\cos\theta)\vec{v}$:

$$A\cos(j\theta) = (2 - 2\cos\theta)\cos(j\theta), \qquad A\sin(j\theta) = (2 - 2\cos\theta)\sin(j\theta)$$

One calculation does the work of two.

Boundary nodes $j = 0, j = N + 1$: $\cos 0 = 1$ fails, $\sin 0 = 0$ passes; $\sin\big((N+1)\theta\big) = 0$ gives $\theta = m\pi/(N+1)$, as in step 3.

> **Result.** With $e^{i\varphi}$, a shift or a derivative multiplies a wave by a factor. For the model matrix the factor is the eigenvalue $2 - 2\cos\theta$.

Two grids, two kinds of waves

From now on a **periodic** domain (e.g. a latitude circle): waves are simplest there.

	model matrix	**periodic grid (from now on)**
grid points	$x_j = jh, j = 1, \dots, N$, plus 2 boundary nodes	$x_j = j\Delta x, j = 0, \dots, N-1$, no boundary
waves	sines, m **half** waves, $k = m\pi/L$	cosines and sines, ℓ **whole** waves ($= 2\ell$ half waves), $k = 2\pi\ell/L$
index	$m = 1, \dots, N$	$\ell = 0, 1, \dots, N/2$

What is in a signal?

N temperatures $u_0, \dots, u_{N-1}$ at stations around a latitude circle. **Which waves is the signal made of, and how strong is each?**

- wave ℓ: a pattern that repeats ℓ times around the circle; $\ell = 0$: the mean
- at the grid points $x_j = jL/N$: $k_\ell x_j = \dfrac{2\pi\ell}{L} \cdot \dfrac{jL}{N} = \dfrac{2\pi\ell j}{N}$

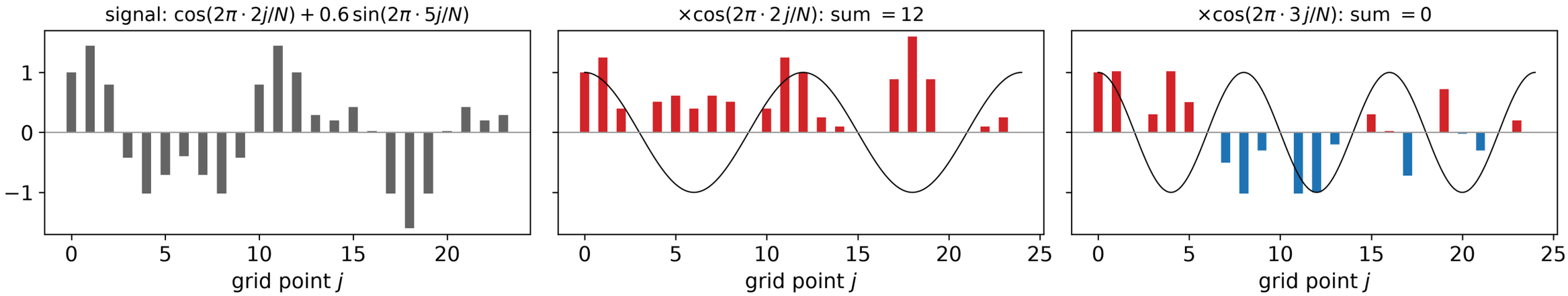
A wave needs two numbers, **amplitude and position**: a cosine and a sine for each ℓ:

$$u_j = a_0 + \sum_{\ell=1}^{N/2} \left[a_\ell \cos \frac{2\pi\ell j}{N} + b_\ell \sin \frac{2\pi\ell j}{N} \right]$$

Up to $\ell = N/2$: N values give at most N coefficients, and $N/2$ is the shortest wave (more under aliasing).

The idea: multiply and sum

Multiply the data point by point with a **test wave**, add up: $\sum_j u_j \cos(2\pi n j/N)$



Think first: why is the sum large for $n = 2$ and zero for $n = 3$?

Matching wave: both have the same sign everywhere, the products add up. Other wave: products of both signs, they cancel.

From the sum to the coefficient

Insert the signal: all other waves drop out, the matching cosine meets itself,

$$\sum_j u_j \cos\frac{2\pi n j}{N} = a_n \sum_j \cos^2\frac{2\pi n j}{N} = a_n \frac{N}{2} \qquad \left(\cos^2\alpha = \tfrac{1}{2}(1+\cos 2\alpha)\text{: average } \tfrac{1}{2}\right)$$

Check with the figure: $N = 24, n = 2$, so the sum should be $a_2 \cdot 24/2 = 12\,a_2$. The figure gives 12, so $a_2 = 1$: the amplitude we built in ✓

$$\boxed{a_n = \frac{2}{N}\sum_{j=0}^{N-1} u_j \cos\frac{2\pi n j}{N}, \qquad b_n = \frac{2}{N}\sum_{j=0}^{N-1} u_j \sin\frac{2\pi n j}{N}}, \qquad 0 < n < \frac{N}{2}$$

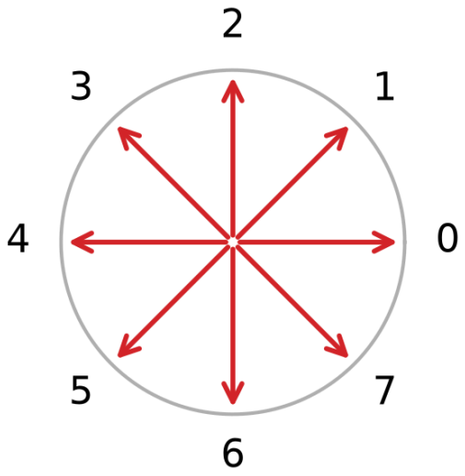
$a_0 = \frac{1}{N}\sum_j u_j$: the mean. This is the **discrete Fourier transform** (DFT).

> Same idea as $\beta_m = \vec{v}_m^T \vec{b}$ for the eigenvectors: the waves are orthogonal, the coefficient is data times wave.

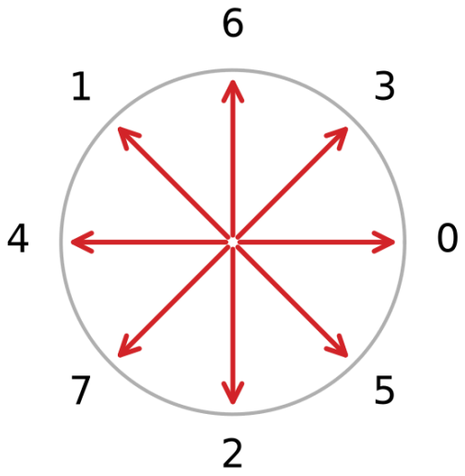
Why the other waves cancel

Product of two waves = waves with $n + \ell$ and $n - \ell$ crests. So: why is **a wave summed over all grid points zero**?

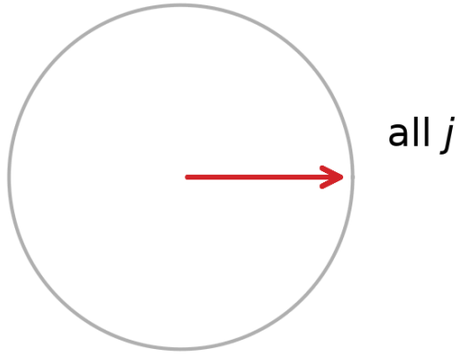
$e^{2\pi i\, 1j/8}$: sum $= 0$



$e^{2\pi i\, 3j/8}$: sum $= 0$



$e^{2\pi i\, 8j/8}$: sum $= 8$



Points spread **evenly** around the circle balance: the cosines (real part) and sines (imaginary part) both sum to zero. Only $p = 0, N, 2N, \ldots$ puts all points at 1: sum N.

The complex form

Pack a_n and b_n into one number, with $e^{-i\alpha} = \cos\alpha - i\sin\alpha$:

$$\boxed{\hat{u}_n = \frac{1}{N}\sum_{j=0}^{N-1} u_j\, e^{-2\pi i\, nj/N}} = \frac{a_n - i\, b_n}{2}$$

- Nothing new: real part = cosine coefficient, imaginary part = sine coefficient
- Amplitude $\sqrt{a_n^2 + b_n^2} = 2|\hat{u}_n|$; angle of $\hat{u}_n$ = position of the crest
- Factor 2: a real wave is two rotations $e^{\pm i\alpha}$; the other half sits in $\hat{u}_{N-n}$

Back from the amplitudes, the **inverse DFT**:

$$u_j = \sum_{\ell=0}^{N-1} \hat{u}_\ell\, e^{2\pi i\, \ell j/N}$$

A worked example

$N = 4,\ u = (1, 2, 1, 0)$. Test waves $n = 1$: $\cos(\pi j/2) = (1, 0, -1, 0),\ \ \sin(\pi j/2) = (0, 1, 0, -1)$.

Think first: multiply and sum. What are a_0, a_1, b_1?

$$a_0 = \tfrac{1}{4}(1 + 2 + 1 + 0) = 1, \qquad a_1 = \tfrac{2}{4}(1 - 1) = 0, \qquad b_1 = \tfrac{2}{4} \cdot 2 = 1$$

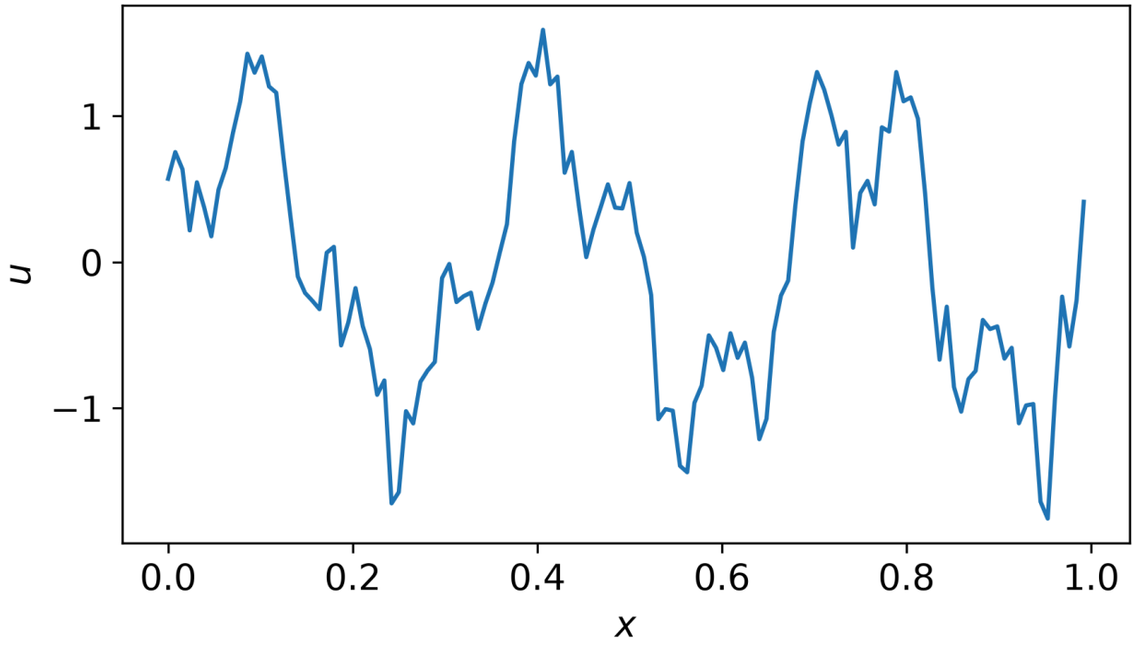
So $u_j = 1 + \sin(\pi j/2)$ ✓, and $\hat{u}_1 = (0 - i)/2 = -0.5i$, amplitude $2|\hat{u}_1| = 1$.

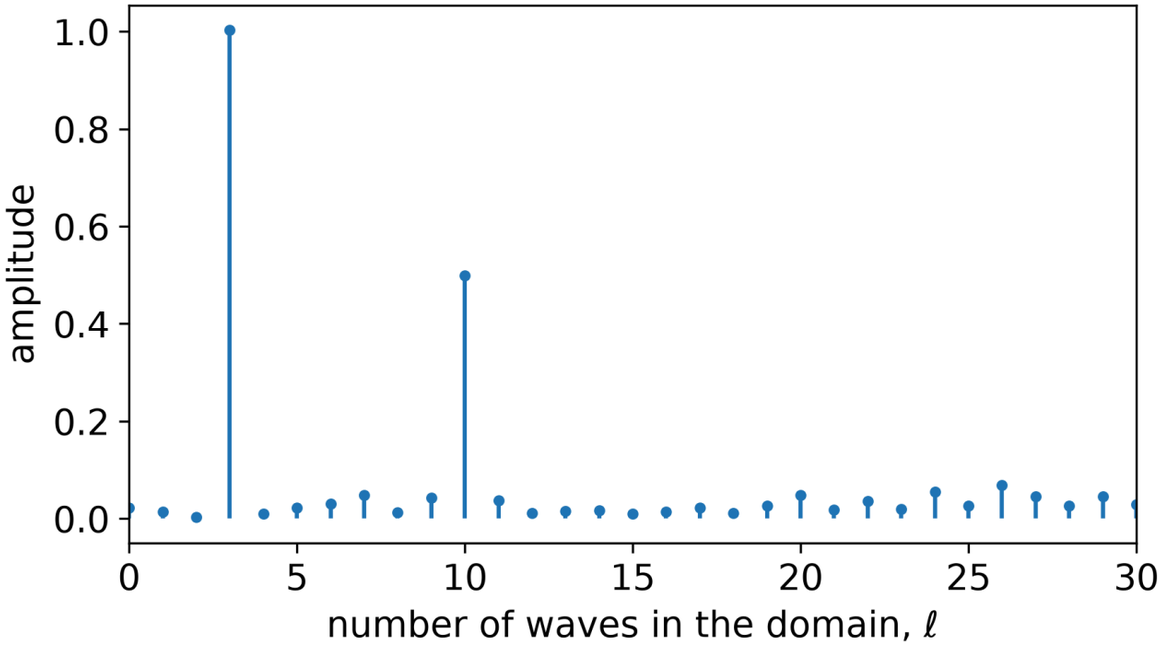
```
u = np.array([1.0, 2.0, 1.0, 0.0])
u_hat = np.fft.rfft(u) / len(u)     # rfft returns N * u_hat, n = 0, ..., N/2
print(u_hat, 2 * abs(u_hat[1]))
```

```
[1.+0.j  0.-0.5j 0.+0.j ] 1.0
```

FFT: $O(N \log N)$ instead of N^2 operations. Amplitude spectrum: `2 * abs(np.fft.rfft(u)) / N`.

A spectrum





$\sin(2\pi \cdot 3x) + 0.5\cos(2\pi \cdot 10x)$ + noise: two peaks, amplitudes 1 and 0.5, on a low noise floor.

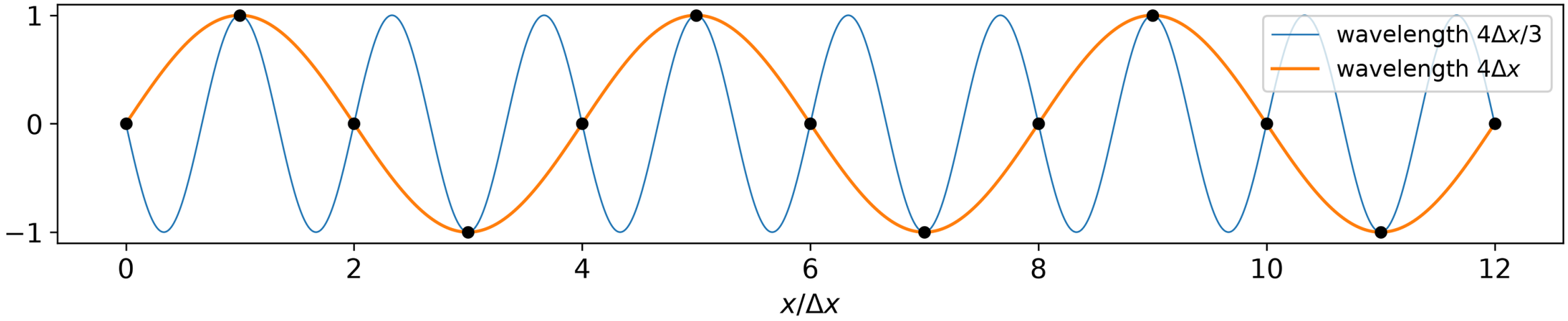
The shortest wave and aliasing

Shortest wave: crest and trough on neighbouring points, wavelength $2\Delta x$, **Nyquist wavenumber** $k_{\max} = \pi/\Delta x$.

Think first: what does the grid make of a wave shorter than $2\Delta x$?

$$e^{i(k+2\pi/\Delta x)\, x_j} = e^{ikx_j}\, e^{2\pi i\, j} = e^{ikx_j}$$

Same values at all grid points: it is mistaken for a longer wave, **aliasing**. Shorter waves do not vanish, they appear as longer ones.



Aliasing in nonlinear models

$$e^{ik_1 x}\, e^{ik_2 x} = e^{i(k_1+k_2)x}$$

- A nonlinear term such as $u\,\partial u/\partial x$ creates **shorter** waves from longer ones
- If $k_1 + k_2 > k_{\max}$, the grid shows the product as a longer wave
- Energy appears at scales where it does not belong: a source of instability in nonlinear models

Waves as a test for numerical methods

The modified wavenumber

Linear equation, constant coefficients: each wave evolves on its own. Exact derivative: $\frac{\mathrm{d}}{\mathrm{d}x} e^{ikx} = ik\, e^{ikx}$.

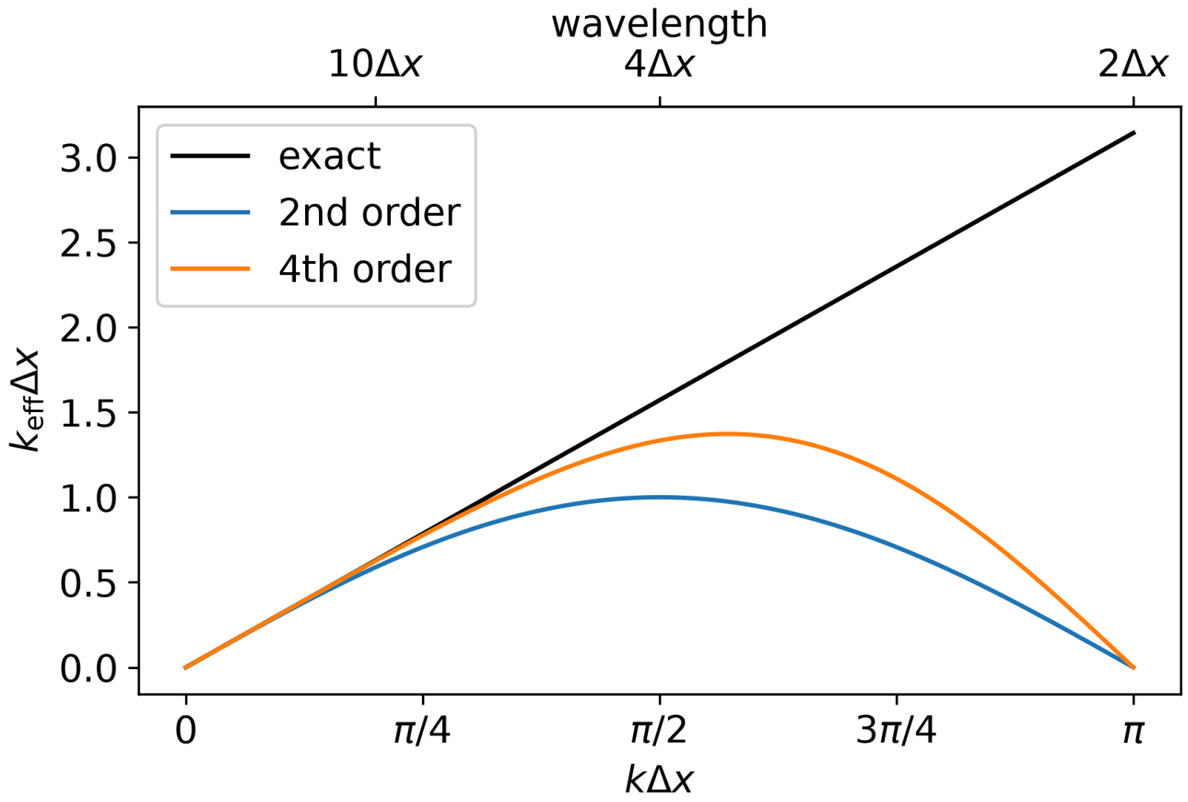
Central difference, factor out e^{ikx}, use $e^{i\alpha} - e^{-i\alpha} = 2i \sin\alpha$:

$$\frac{e^{ik(x+\Delta x)} - e^{ik(x-\Delta x)}}{2\Delta x} = \frac{e^{ik\Delta x} - e^{-ik\Delta x}}{2\Delta x}\, e^{ikx} = i\, \frac{\sin(k\Delta x)}{\Delta x}\, e^{ikx}$$

$$k_{\mathrm{eff}} = \frac{\sin(k\Delta x)}{\Delta x} \approx k \left[1 - \frac{(k\Delta x)^2}{6}\right]$$

Relative error $(k\Delta x)^2/6$: the truncation error, **wave by wave.**

What "second order" hides



- Long waves: excellent
- $4\Delta x$ wave: derivative too small by $\sin(\pi/2)/(\pi/2) = 0.64$
- $2\Delta x$ wave: derivative **zero**, not seen at all
- About **10 points per wavelength** for 6 % error

Effective resolution: a weather model with 2 km grid spacing resolves features of about 10 km.

Two views of one fact

Second difference of a wave, with $e^{i\alpha} + e^{-i\alpha} = 2\cos\alpha$:

$$\frac{e^{ik(x+\Delta x)} - 2e^{ikx} + e^{ik(x-\Delta x)}}{\Delta x^2} = -\frac{2 - 2\cos(k\Delta x)}{\Delta x^2}\, e^{ikx} = -\frac{4}{\Delta x^2}\sin^2\frac{k\Delta x}{2}\, e^{ikx}$$

Exact: $-k^2$. Long waves, $\sin\alpha \approx \alpha$: again about $-k^2$.

Model matrix: $\Delta x = h$, $k_m = m\pi/L$, so $k_m\Delta x = \theta_m$:

$$-\frac{4}{h^2}\sin^2\frac{\theta_m}{2} = -\frac{\lambda_m}{h^2} \qquad (\text{second difference} = -A/h^2)$$

Result. Waves are the eigenvectors of difference operators with constant coefficients; the factor is the eigenvalue. The wave analysis gives λ_m without the matrix. For a full scheme in space and time: **von Neumann stability analysis**.

Summary

- **Model matrix**: $-u'' = f \to A\vec{x} = \vec{b}, A = \text{tridiag}(-1, 2, -1), \vec{b} = h^2\vec{f}$; sparse, SPD → exactly one solution
- **Eigenvalues**: $\lambda_m = 4\sin^2\left(m\pi/(2(N+1))\right), m = 1, \dots, N$ half waves, sine eigenvectors; $\lambda_1 \approx \pi^2/(N+1)^2, \lambda_N \approx 4$
- **Norm, condition**: $\|A\|_2 = \max|\lambda_m|$; $\kappa = \lambda_{\max}/\lambda_{\min} \approx 4(N+1)^2/\pi^2 \sim 1/h^2$
- **Waves on a grid**: DFT splits N values into N waves; shortest $2\Delta x$, shorter ones aliased
- **Difference formulas**: multiply a wave by a factor; $k_{\text{eff}} = \sin(k\Delta x)/\Delta x$ → ~10 points per wavelength; second difference: $-\lambda_m/h^2$

Lab and next lecture

- **Lab**: Exercise 4 (the model matrix, its eigenvalues and condition number), back next week
- Add the model matrix to your `toolbox.py`
- **Reading**: script, unit 4
- **Next lecture**: one-step methods for ordinary differential equations